

ASTU University Analytics Report

Adama Science and Technology University
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Executive Summary

Total Projects: 22
Research Projects: 15
Community Projects: 7
Active Colleges: 6
Total Funding (ETB): 19,733,000
Total Publications: 52
People Benefited: 20,040
Volunteers Engaged: 340

Project Status Breakdown

Under_review: 2 projects
Paused: 1 projects
Active: 15 projects
Completed: 3 projects
Planned: 1 projects

Projects by College

College of Electrical Engineering & Computing: 7
College of Applied Natural Science: 4
College of Mechanical, Chemical & Materials Engineering: 3
College of Engineering: 3
College of Medical and Health Sciences: 2
College of Business and Economics: 2
College of Civil Engineering and Architecture: 1

Research Projects

1. High-Temperature Superconductivity in Nanostructured Fe-Based Alloys

Lead: Dr. Solomon Bekele | College: College of Mechanical, Chemical & Materials Engineering | Status: under_review | Funding: ETB 1,450,000

Investigating structural alloys under extreme thermal stresses to measure transition temperatures.

2. Automated Micro-Grid Energy Balancing System for Off-Grid Campus

Lead: Dr. Robel Tadesse | College: College of Electrical Engineering & Computing | Status: under_review | Funding: ETB 1,850,000

Designing control algorithms for solar hybrid microgrids on campus.

3. Amharic Speech-to-Text Recognition System using Transformers

Lead: Dr. Hana Tesfaye | College: College of Electrical Engineering & Computing | Status: paused | Funding: ETB 920,000

Developing open source automatic speech recognition tools for low-resource Amharic dialects.

4. AI-Powered Crop Disease Detection Using Deep Learning

Lead: Dr. Tesfaye Worku | College: College of Electrical Engineering & Computing | Status: active | Funding: ETB 850,000

Developing a mobile-first deep learning system to detect crop diseases from smartphone photos, targeting smallholder farmers in Oromia region.

5. Solar-Powered Water Purification for Rural Ethiopia

Lead: Prof. Almaz Tadesse | College: College of Mechanical, Chemical & Materials Engineering | Status: active | Funding: ETB 1,200,000

Designing low-cost solar-driven water treatment units deployable in off-grid communities, with pilot testing in Afar and SNNP regions.

6. Seismic Risk Assessment of Adama Urban Infrastructure

Lead: Dr. Biruk Hailu | College: College of Civil Engineering and Architecture | Status: active | Funding: ETB 2,100,000

Comprehensive seismic vulnerability mapping of Adama city buildings using GIS, field surveys, and finite element modelling.

7. Blockchain-Based Land Registry System for Ethiopia

Lead: Dr. Yonas Girma | College: College of Electrical Engineering & Computing | Status: active | Funding: ETB 950,000

Prototyping a tamper-proof, decentralised land ownership record system to reduce disputes and corruption in land administration.

8. Teff Genome Sequencing and Nutritional Enhancement

Lead: Prof. Mekdes Bekele | College: College of Applied Natural Science | Status: active | Funding: ETB 1,750,000

Full genome sequencing of 12 teff varieties to identify genes linked to high iron and zinc content for nutritional improvement.

9. Wind Energy Potential Mapping of Ethiopian Rift Valley

Lead: Dr. Solomon Bekele | College: College of Electrical Engineering & Computing | Status: completed | Funding: ETB 680,000

High-resolution wind resource assessment using meteorological stations, satellite data and computational fluid dynamics modelling.

10. Machine Learning for Amharic Natural Language Processing

Lead: Dr. Hana Tesfaye | College: College of Electrical Engineering & Computing | Status: active | Funding: ETB 720,000

Building an open-source Amharic NLP toolkit covering sentiment analysis, named entity recognition and machine translation.

11. Geothermal Energy Exploration in Aluto-Langano

Lead: Prof. Getachew Mengistu | College: College of Applied Natural Science | Status: active | Funding: ETB 3,200,000

Subsurface characterisation and resource estimation of the Aluto-Langano geothermal field using seismic, gravity and MT surveys.

12. Smart Traffic Management System for Adama City

Lead: Dr. Robel Tadesse | College: College of Electrical Engineering & Computing | Status: planned | Funding: ETB 890,000

IoT sensor network and adaptive signal control algorithm to reduce congestion and vehicle emissions in Adama's central districts.

13. Traditional Medicinal Plants of Oromia: Pharmacological Study

Lead: Dr. Chaltu Wakjira | College: College of Applied Natural Science | Status: active | Funding: ETB 560,000
Systematic screening of 45 Oromo traditional medicinal plants for antimicrobial, anti-inflammatory and antidiabetic properties.

14. Drought Prediction Model Using Satellite Remote Sensing

Lead: Dr. Fikirte Haile | College: College of Applied Natural Science | Status: active | Funding: ETB 980,000
Random forest model integrating MODIS NDVI, CHIRPS rainfall and GRACE groundwater anomaly data to predict drought onset 3 months ahead.

15. Biogas Production from Coffee Processing Waste

Lead: Prof. Dawit Asfaw | College: College of Mechanical, Chemical & Materials Engineering | Status: completed
| Funding: ETB 430,000
Anaerobic digestion optimisation of coffee pulp and wastewater to produce biogas for rural household energy, with field pilots in Jimma.

Community Projects

1. Digital Literacy for Secondary School Teachers

Lead: Dr. Hana Tesfaye | Location: East Shewa Zone | Beneficiaries: 480 | Status: completed
Intensive 3-week ICT training for 480 secondary school teachers covering office software, internet research and basic data management.

2. Mental Health Awareness on ASTU Campus

Lead: Dr. Mekdes Bekele | Location: ASTU Campus | Beneficiaries: 6,500 | Status: active
Peer counselling network, monthly awareness seminars and anonymous helpline supporting student mental health across all departments.

3. Micro-Enterprise Support for Adama Street Vendors

Lead: Prof. Tesfaye Demissie | Location: Adama City Markets | Beneficiaries: 760 | Status: active
Business registration support, bookkeeping training and group savings scheme for informal street vendors in Adama city centre.

4. Clean Water Access for Wolenchiti Kebele

Lead: Prof. Almaz Tadesse | Location: Wolenchiti, East Shewa | Beneficiaries: 2,400 | Status: active
Installation of solar-powered water pumping system and distribution network serving 2,400 residents of Wolenchiti kebele.

5. Women Entrepreneurship & Financial Literacy Programme

Lead: Dr. Selamawit Girma | Location: Adama & Modjo | Beneficiaries: 580 | Status: active
Business skills training, micro-loan facilitation and mentorship for women-owned small enterprises in Adama and Modjo towns.

6. ASTU Blood Donation Campaign

Lead: Dr. Chaltu Wakjira | Location: Adama, Multiple Sites | Beneficiaries: 9,000 | Status: active
Bi-monthly blood donation drives across ASTU campus and partner high schools, supplying Adama General Hospital and nearby health centres.

7. Free Coding Bootcamp for Adama Youth

Lead: Dr. Robel Tadesse | Location: Adama City | Beneficiaries: 320 | Status: active
12-week intensive programming bootcamp for unemployed youth aged 18-30 in Adama, covering Python, web development and mobile apps.